DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2022

Course: B. Tech. **Branch : Chemical** **Semester :VI**

Subject Code & Name: BTCHE 606A Catalyst Science and Technology

Max Marks: 60 **Date:29/08/2022** **Duration: 3.45 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/ CO)	Marks
Q. 1	**Solve Any Two of the following.**		
A)	**What are the basic steps in heterogeneous catalysis process?**	**CO1**	**6**
B)	**Define adsorption isotherm? Derive Langmuir equation. State the assumptions, if any.**	**CO1**	**6**
C)	**Distinguish between physisorption and chemisorption?**	**CO1**	**6**
Q.2	**Solve Any Two of the following.**		
A)	**What are the major components of a catalyst? What is the function of major component in a solid catalyst, explain in brief.**	**CO3**	**6**
B)	**What are the major mechanistic steps in precipitation process? What is the effect of process parameters on product properties, Explain in brief?**	**CO1**	**6**
C)	**Why catalyst formulation and shaping is needed? Explain the various formulation techniques in brief.**	**CO1** **CO1**	**6**
Q. 3	**Solve Any Two of the following.**		
A)	**Why this characterization is important for catalyst. List the different characterization techniques?.**	**CO2**	**6**
B)	**Define magnification and resolution of a microscope. What are the advantages of electron microscope over optical microscope?**	**CO2**	**6**
C)	**Differentiate in infrared and Raman spectroscopy?.**	**CO2**	**6**
Q.4	**Solve Any Three of the following.**		
A)	**What are the main steps to prepare supported catalyst? Explain the dry and wet impregnation method in brief.**	**CO1**	**6**
B)	**Define combine and effective diffusivity? Which models are commonly used to determine diffusivity?**	**CO1**	**6**

C)	Determine the effect of intraparticle mass transfer resistance on the order of a reaction	CO2	6

Q. 5 Solve Any THREE of the following.

A)	What are the different ways of catalyst deactivation? Explain in brief.	CO3	4
B)	Which type of reactor is used in hydrocracking process? State the limitations associated. Also draw the schematic of two stage hydrocracking process.	CO4	4
C)	What is monolithic catalyst? Explain in brief with example.	CO4	4
D	What are the heterogeneous catalytic processes involve mainly in petroleum refining industries.	CO4	4

*** End ***